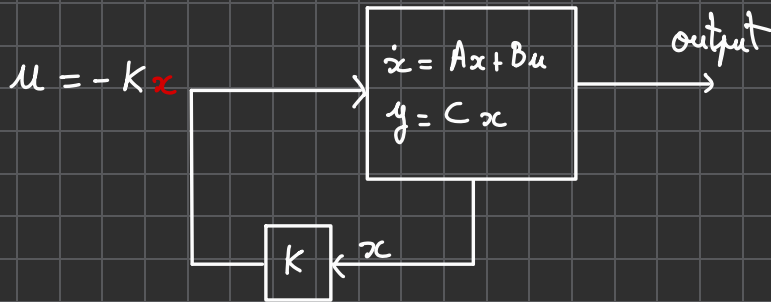


03) State Observer.

A State observer is a mathematical tool used to estimate the internal states of a dynamic system based on its input/output and a known system model.

For deciding input value by $u = -Kx$, it is required to know value x for all time



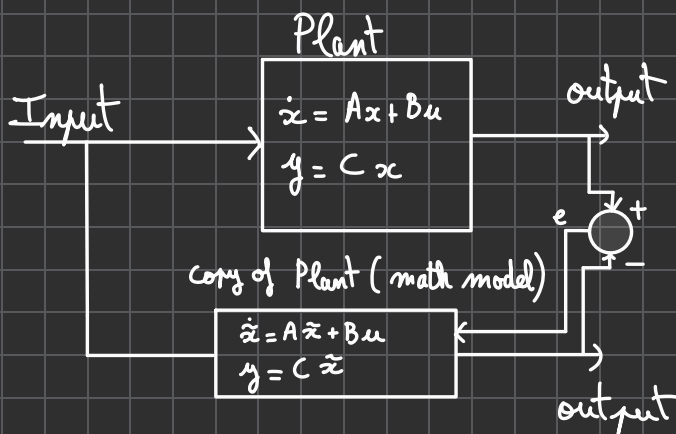
In case of observable or detectable.

→ State of the plant can be estimated by using "State observer"

When rank of observability matrix $\mathcal{O}_0$ is n , the plant is observable

$$\mathcal{O}_0 = \begin{bmatrix} [C] \\ [CA] \\ \vdots \\ [CA^{n-1}] \end{bmatrix}$$

Concept of "State Observer"



State Observer for continuous-time system

State Observer:

$$\dot{\tilde{x}} = (A - LC) \tilde{x} + \overset{\text{observer gain}}{L} y + Bu$$

Estimated Error: $e = x - \tilde{x}$

$$\dot{e} = (A - LC) e$$

If no model error or disturbance, following equation holds

L is settle to stabilize system

$$\Rightarrow \lim_{t \rightarrow \infty} e(t) = 0$$

State Observer for discrete-time system

State Observer:

$$\tilde{x}(k+1) = (A - LC) \tilde{x}(k) + L y(k) + Bu(k)$$

Estimated Error: $e = x - \tilde{x}$

$$e(k+1) = (A - LC) e(k)$$

If no model error or disturbance, following equation holds

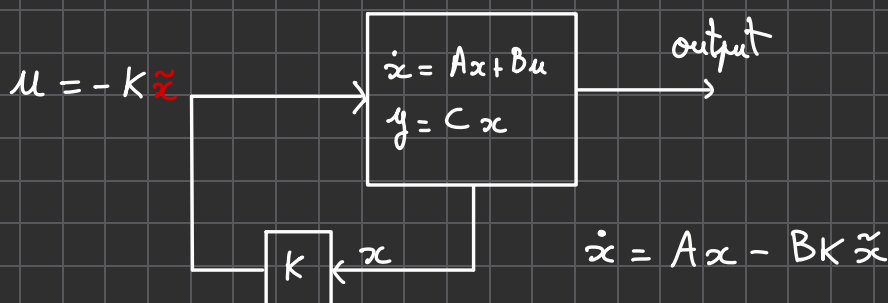
L is settle to stabilize system

$$\Rightarrow \lim_{t \rightarrow \infty} e(t) = 0$$

Poles of $(A - LC)$ are in the unit circle (Stabilize discrete time system)

Observer-based control system

- State feedback control, **Estimated state** is used instead of the state.



- State feedback $\tilde{x} = x - e$

$$\dot{x} = Ax - Bk \tilde{x}$$

- State Observer

$$\dot{\tilde{x}} = (A - Lc) \tilde{x} + Ly + Bu$$

$$\Rightarrow \dot{\tilde{x}} = (A - Lc) \tilde{x} + Lc x - Bk \tilde{x} \quad (u = -K \tilde{x})$$

- Estimated error

$$\dot{e} = (A - Lc) e$$

Case 1: State feedback + State observer

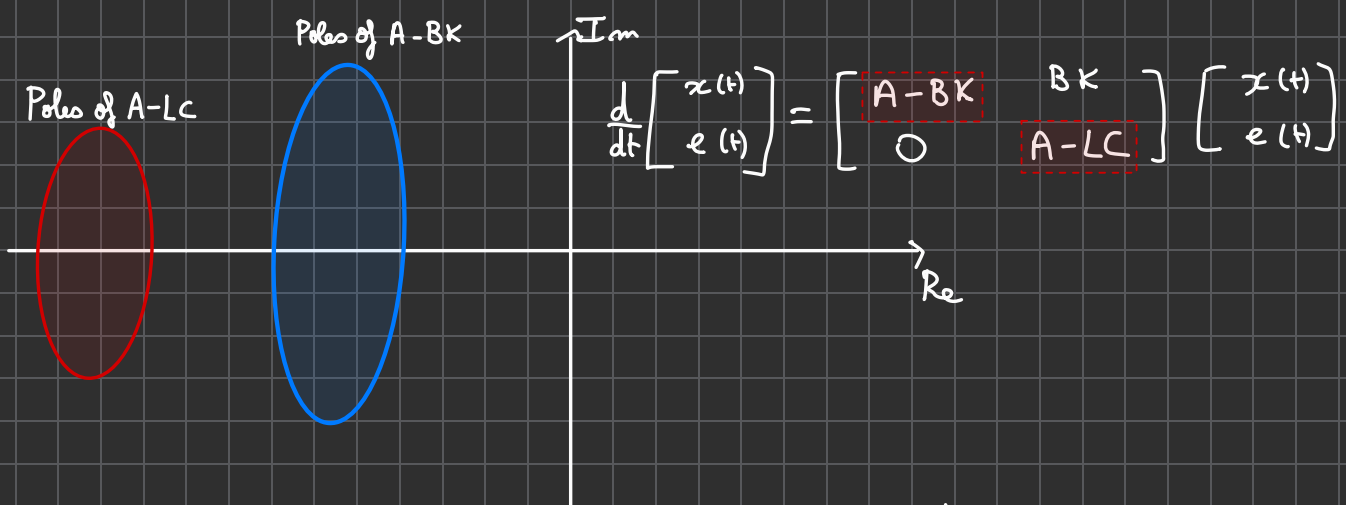
$\Rightarrow$ observer based control system

$$\frac{d}{dt} \begin{bmatrix} x(t) \\ \tilde{x}(t) \end{bmatrix} = \begin{bmatrix} A & -BK \\ LC & A - LC - BK \end{bmatrix} \begin{bmatrix} x(t) \\ \tilde{x}(t) \end{bmatrix}$$

Case 2: State feedback + Estimated error.

observer based control system

$$\Rightarrow \frac{d}{dt} \begin{bmatrix} x(t) \\ e(t) \end{bmatrix} = \begin{bmatrix} A - BK & BK \\ 0 & A - LC \end{bmatrix} \begin{bmatrix} x(t) \\ e(t) \end{bmatrix}$$



for achieving dynamics by $(A-BK)$ for observer-based control, it is required to assign poles of $(A-LC)$ as far from imaginary axis compared to $(A-BK)$